

# Concept Document for *Overcooked!* Tabletop

Group Project #2 for GSND 5110, Group #6

[Trello Board](#)

[Team Log](#)

## Group Members

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## Overview

先简要介绍原始游戏和改编版本的概况。我写。

## Original Game Dissection

剖析原始游戏之“essense”。把第一周时我写的那一坨粘进来即可。

## Mechanism Design

承继上节，介绍我们是如何将那些 essense 改编成桌游机制的。柴哥写。

## MDA Analysis

将上节总结成表。亦驰写。

## Difficulty Model

### Identifying the Parameters

The goal of the game is to get as many scores as possible by finishing orders before the rounds run out. This means that the scores the players get from the orders is one target parameter of the game. Because there is little random factors throughout the game process, we state that the game's difficulty could be measured from the design of the map and the orders.

In this prototype, there are these key parameters:

- Map design: The position of each facilitation, and the distances to move between them.

- Orders: The type of the orders, the scores of each type, the time limitation of each order and the amount of the orders.
- Mechanism: The movement per round per player  $M$ , and the scores required to pass the level  $S$ .

## Moving Strategies

For each ingredient component of an order, there is theoretically a minimal movement to complete cooking it. From this, we could estimate the minimal required movement for each order, and use that as the foundation for the numerical design of the orders.

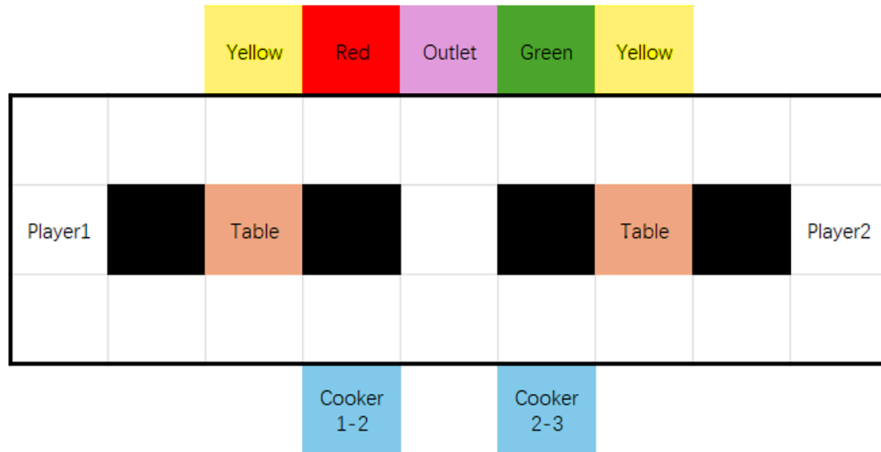


Figure 1: The map layout of the game.

Take the map in Figure 1 as an example. Suppose that a player starts in front of *Outlet*, to fetch the *yellow* ingredient and cook it to level 3, the necessary movement could be:

$$\begin{aligned}
 & 3(\text{Get ingredient}) \\
 & + 6(\text{To cooker 1-2}) + 1(\text{Cook}) \\
 & + 3(\text{To cooker 2-3}) + 1(\text{Cook}) \\
 & + 3(\text{To outlet}) + 1(\text{Serve}) \\
 & = 18.
 \end{aligned}$$

Similarly, we could calculate the estimated minimal movement for any orders.

However, considering that the main aesthetic of the game is the cooperation between players, we think that it is mainly reflected on two points:

1. Use the *table* block provided in the game to pass the ingredients and avoid unnecessary movement.
2. Avoid running into each other while moving as it leads to waste of movement.

Therefore it is also necessary to calculate the minimal movement when the *table* blocks are used:

| Player 1                                  | Player 2                                  |
|-------------------------------------------|-------------------------------------------|
| 3(Get ingredient)                         | 1(Take ingredient from table)             |
| + 1(Place on table)                       | + 2(To cooker 1-2) + 1(Cook)              |
| + 4(Take and serve the cooked ingredient) | + 3(To cooker 2-3) + 1(Cook)              |
| = 8.                                      | + 4(Place the cooked ingredient on table) |
|                                           | = 12.                                     |

Speaking from the numbers, when not using table, every player could finish cooking a level-3 ingredient within a minimal movement of 18; but when using table, the number would increase to 20. Albeit not precise, it seems like the overall effectiveness would be higher if the players choose not to use the table.

But this is not the reality. As said previously, there is punishment for collisions in our game: Whenever players are moving into the same square, they would collide due to crashing into each other, causing the movement to fail and all remaining actions for this turn to be lost. Because it is hard to predict players' actions, we assume that the probability of the occurrence per round per action is constant; then the loss of movement per collision could be given as Equation 1.

$$2 \times \left( \frac{M-1}{2} + 1 \right). \quad 1$$

In our current design,  $M = 7$ ; so the loss of movement per collision would be  $2 \times \left( \frac{7-1}{2} + 1 \right) = 8$ .

It now can be seen that as long as collision happens, the expected total movement for cooking one ingredient with a non-cooperative strategy would be higher than a cooperative strategy, because in cooperative plays there is no chance for the players to collide. The three squares on the middle axis of the map would be passed by the players the most frequently, also the order of using the cookers is fixed, so the chance of a collision happening is significantly high.

It is certainly possible to use other strategies than cooperation to avoid collisions. For example, one player could always use the outer route to go to the cooker, so that it would take 4 more movements to cook an ingredient than the optimal route, resulting in the same overall performance with a cooperative strategy; but this strategy also cannot avoid collisions completely, as there is only one cooker of each kind on the map, and it will take one around to do the cooking. The players themselves would become the blockades on the path.

## Level Difficulty

Here we will discuss in detail about another non-intuitive yet key parameter: The minimal required movement to prepare all ingredients in an order. We hope that the score of each order could proportionally match their minimal movement to avoid the imbalance between different kinds of order. We established a standard ratio  $Q$  of the score to the expected movement of an order, and managed to make sure that this statistic of all orders' are close to  $Q$  (they don't have to be exactly equal).

The prototype mainly tests the cooperation level between the players and their ability to plan the moving routes. Either insufficient cooperation, adopting a non-optimal strategy and colliding with other players would lead to waste of movement compared to the optimal strategy. To help understand, we have the following formulae:

$$\begin{array}{ll}
\#(\text{actual movement to win}) & \#(\text{total available movement}) \\
- \#(\text{theoretical minimum movement}) & - \#(\text{theoretical minimum movement}) \\
= \#(\text{movement wasted}), & = \#(\text{movement that could be wasted}).
\end{array}$$

When the movement wasted is more than the movement that could be wasted, the rest available movement could be less than the movement required to win the game, meaning that the players would likely lose the game. The closer the theoretical minimum movement to the total available movement, the more difficult it is for the player to win the game. We define the level difficulty to be the ratio of these two: (Equation 2)

$$D = \frac{\#(\text{theoretical minimum movement})}{\#(\text{total available movement})}. \quad 2$$

By investigating the order pile, some parameters could be estimated. Until the order pile is emptied, there are always 3 active orders, which means that the order timer would decrease by 3 each round. Then the total available movement could be estimated as Equation 3.

$$(\text{total available movement}) = \frac{1}{3} \sum_{\text{order pile}} (\text{order time}) \times M \times 2(\# \text{player}). \quad 3$$

Also, the theoretical minimum movement is given by Equation 4.

$$(\text{theoretical min. movement}) = \frac{(\text{required score to win})}{Q}. \quad 4$$

When a level design is finished, the expected movement for finishing each order is then decided. By adjusting the amount and the type of the orders in the pile, and their score and time limitation, we could adjust the players' total available movement freely, as well as the theoretical minimum movement required by the level. This means that only by adjusting the order pile and the level passing score, we could control the level difficulty.

## Adjusting

We hope that in the first level,  $Q = 60\%$ , while in the second level  $Q = 78\%$ . Numerically, this meets the expectation that the second level would be 30% harder than the first level; but it still remains a question whether this is what the players would feel about the difficulty, which needs to be find out by further playtesting.

Below are the numerical analysis of this map.

## Individually Optimal Strategy

Let's assume that there is only one player who spawns below the *Outlet* square and returns to the same square after each order is made. The expected movement for cooking each type of ingredient would be given as Table 1.

| Type    | Movement |
|---------|----------|
| Yellow2 | 13       |
| Yellow3 | 17       |
| Red2    | 12       |
| Red3    | 16       |
| Green2  | 12       |
| Green3  | 16       |

Table 1: The expected movement for cooking ingredients in the IO strategy.

Back to the two-player case, we let the expectation of the amount of collision occurrence within one cooking process be  $K_1$ .

### Longer Route Strategy

In this strategy, to avoid collision, one player would take longer routes by the rim of the map, but still returns to the origin from the middle passageway after cooking. Then the expected movement for each ingredient would be given as Table 2.

| Type    | Movement |
|---------|----------|
| Yellow2 | 16       |
| Yellow3 | 20       |
| Red2    | 16       |
| Red3    | 20       |
| Green2  | 18       |
| Green3  | 22       |

Table 2: The expected movement for cooking ingredients in the LR strategy.

We also let  $K_2$  be the same expectation of collision under this strategy.

### Cooperative Strategy

In this strategy, one player always stays in the upper half of the board, passes ingredients to the tables, and take cooked ingredients from the table and serve them; the other player always stays in the lower half of the board and cook the passed ingredients. The expectation table would be Table 3.

| Type    | Player 1 | Player 2 |
|---------|----------|----------|
| Yellow2 | 8        | 6        |
| Yellow3 | 8        | 12       |
| Red2    | 8        | 6        |
| Red3    | 8        | 12       |
| Green2  | 8        | 6        |
| Green3  | 8        | 12       |

Table 3: The expected movement for cooking ingredients in the cooperative strategy.

Considering the  $K$  values, the overall table comparing all strategies is shown in Table 4.

| Type    | IO          | LR                                 | Cooperative |
|---------|-------------|------------------------------------|-------------|
| Yellow2 | $13 + 4K_1$ | $(16 + 13)/2 + 4K_2 = 14.5 + 4K_2$ | 14          |
| Yellow3 | $17 + 4K_1$ | $(20 + 17)/2 + 4K_2 = 18.5 + 4K_2$ | 20          |
| Red2    | $12 + 4K_1$ | $(16 + 12)/2 + 4K_2 = 14.0 + 4K_2$ | 14          |
| Red3    | $16 + 4K_1$ | $(16 + 20)/2 + 4K_2 = 18.0 + 4K_2$ | 20          |
| Green2  | $12 + 4K_1$ | $(12 + 18)/2 + 4K_2 = 15.0 + 4K_2$ | 14          |
| Green3  | $16 + 4K_1$ | $(22 + 16)/2 + 4K_2 = 19.0 + 4K_2$ | 20          |

Table 4: Comparison between all strategies.

We can see from the data that, theoretically speaking, regardless of the type of the ingredients, as long as there happens just one collision, the cooperative strategy will become the optimal approach. Via physical testing of the moving strategy, we found that the non-cooperative strategies would cause great waste of movement and waitings. They are most ineffective compared to the cooperative strategy. Based on this, we set the expected movement data under the cooperative strategy to be the standard expectation.

### Orders

We have designed the following orders as shown in Table 5 ( $Q = 1/6$ ).

| Ingredients             | Movement | Scores | Time limit |
|-------------------------|----------|--------|------------|
| Yellow2 + Red2 + Green2 | 42       | 7      | 3          |
| Yellow2 + Red3 + Green3 | 54       | 9      | 4          |
| Yellow3 + Red3 + Green3 | 60       | 10     | 5          |

Table 5: Order designs.

Let there be 5 of each type of order in the pile, then the total number of turns would be  $5 \times (3 + 4 + 5)/3 = 20$ . When  $M = 7$ , the total movememnt would be  $2 \times 20 \times 7 = 280$  steps.

To match the difficulty of level 1, the expected movement should be  $60\% \times 280 = 168$  steps. Given that  $Q = 1/6$ , the required score to pass level 1 should be  $168 \times 1/6 = 28$ .

If the order pile is not changed, then to match the difficulty of level 2, the required passing score should be  $280 \times 78\% \times 1/6 = 36.4$ , rounded down to 36.

## Implementation

We simplify different food into different colors and the cooking state of food into numerical values. As color is an important indicator in this game, expect colorful target items like food, cookers, tables and outlet, all other elements are colorless. The floor is white, the wall and obstacles are black, the edges of squares are gray, and the two players are white. Moreover, only 6 basic colors are used, including red, yellow, green, orange, blue and purple, of which red, yellow and green are used to indicate food related things, including food ingredients and food storage, orange blocks in map are table, blue blocks in map are cookers, and purple block is the outlet of food delivery. The minimalist design of the game is simple and practical to players.

## The Map

As shown in the figure below, the map includes the player activity area and the surrounding equipment placement area. The player activity area is a  $3 \times 9$  checkboard with brown tables and black obstacles (Figure 1). Players can put or pick food on tables, but players cannot pass through tables or black obstacles. Players initial place is shown in squares of Player1 born", "Player2 born". Food storages, cookers, and delivery ports are placed around the player activity area. There are 4 food storages, where players take level 1 red, green, and yellow ingredients respectively. There are 2 cookers, one can process level 1 food to level 2 food, and the other can process level 2 food to level 3 food. There is also a food delivery port middle-upper to the player activity area.

## Orders

As shown in Figure 2, the orders includes two aspects. The left part shows the composition of the menu, and white numbers on the fan-shaped color blocks indicate the degree of cooking of the food. For example, the order below is composed of a level 3 red, a level 2 yellow, and a 3 marked green.

The right side of the menu is the score of this menu and the time limit for order completion. For example, the order below needs to be completed within 3 turns, and the player can get 9 points after completion.

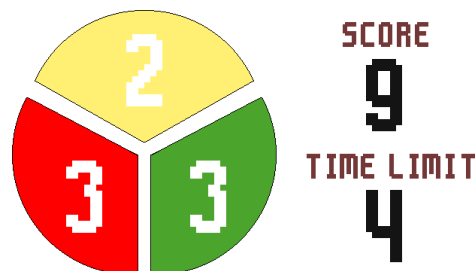


Figure 2: The illustration of an order in game.

## Players

As shown in Figure 3, the 2 players are designed to be colorless and the marked numbers indicate Player 1 and Player 2.



Figure 3: The pieces used to represent the players.

## Food

Different colored food is represented by dice of corresponding color (Figure 4). When food is picked up of the storage, it is placed with 1 point facing up. Food with 1 point facing up can be cooked by cooker 1–2, and after cooking it would be placed with 2 points facing up. Food with 2 points facing up can be cooked by cooker 2–3, and after cooking it would be placed with 3 points facing up.

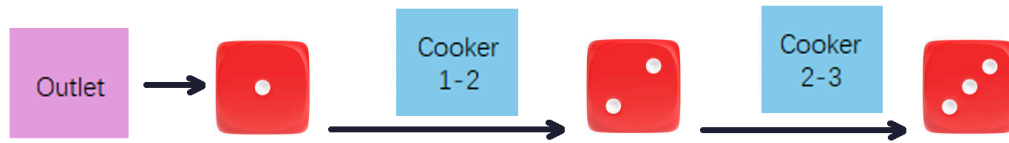


Figure 4: Dice are used to represent food at different stages.

## Playtest and Feedback

测试反馈结果。亦驰写。

## Acknowledgement

Special thanks to *Yichi*'s roommate and our great classmate, *Zhuowen Song* (宋卓文)'s generous effort on playtesting the game.